BioHackathon series:
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Project 9

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Evolving FAIR Image Analysis in Galaxy for Cross-domain and AI-ready Applications

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Introduction

The growing integration of image-based technologies across life sciences, environmental research, and other domains has intensified the need for interoperable, reproducible, and FAIR-compliant image analysis infrastructures. The ELIXIR BioHackathon Europe 2025 provided a collaborative environment to advance these goals within the Galaxy ecosystem (Galaxy Community, 2024).

Previous projects at the ELIXIR BioHackathon Europe focused on community development and establishing conventions (Kostrykin et al., 2024) or implementing FAIR workflows and training materials (Kostrykin et al., 2025) for image analysis using Galaxy. Building on the achievements of the previous years, Project 9 at the ELIXIR BioHackathon Europe 2025 was entitled *"Evolving FAIR Image Analysis in Galaxy for Cross-domain and AI-ready Applications"* and aimed to strengthen Galaxy's capabilities for bioimage analysis while enhancing cross-domain interoperability and reproducibility. The project addressed three major challenges:

Meeting information

If you want to submit a preprint to BioHackrXiv, first check if your meeting is registered. You can find a list of meetings here. If your meeting is missing, please contact your meeting organizers. The above list also provides information on the YAML fields with information about the meeting.

The following fields need to be given:

```
biohackathon_name: "BioHackathon Europe 2023"
biohackathon_url:   "https://biohackathon-europe.org/"
biohackathon_location: "Barcelona, Spain, 2023"
group: Project 26
git_url: https://github.com/yourOrganization/your_report_repo
```

The BioHackrXiv meeting pages provide content to use for the first three fields. The `git_url:` field must have the link to the GitHub repository with your preprint (draft).

Author information

Information about the authors is given in the YAML format at the top of this template. For authors you provide their names, their affiliations. That is the minimum, but as BioHackrXiv is moving to a situation where more metadata is shared, and used by, for example, EuropePMC, adding additional information ie encouraged.

BioHackathons is about hacking together, and the minimal number of authors for reports is two. This makes a minimal example look like this:

```yaml
authors:
  - name: First Author
    affiliation: 1
  - name: Last Author
    affiliation: 2
affiliations:
  - name: First Affiliation
    index: 1
  - name: ELIXIR Europe
    index: 2
```

Author identifiers

Ideally, authors provide their ORCID identifier. For affiliations, It is added with the `orcid:` field. So, and author record would look like this:

```yaml
authors:
  - name: First Author
    affiliation: 1
    orcid: 0000-0000-0000-0000
```

Research Organization Registry identifiers

Matching the author identifier, the affiliations can be further specified with the Research Organization Registry (ROR) identifier. For example, this is the affiliation identifier can be added with the `ror:` field:

```yaml
affiliations:
  - name: ELIXIR Europe
    ror: 044rwnt51
    index: 2
```

Contributor Role Taxonomy

A last feature since is minimal support for the Contributor Role Taxonomy (CRediT). You can specify the role of authors in writing the report with the `role:` field. However, the authors are responsible for selection the right terms from CRediT. An example looks like this:

```yaml
authors:
  - name: First Author
    affiliation: 1
    orcid: 0000-0000-0000-0000
    role: Conceptualization, Writing - review & editing
```

A full examples

A full example then has this structure:

```yaml
authors:
  - name: First Author
    affiliation: 1
    role: Writing - original draft
  - name: Last Author
    orcid: 0000-0000-0000-0000
    affiliation: 2
```

```
    role: Conceptualization, Writing - review & editing
affiliations:
  - name: First Affiliation
    index: 1
  - name: ELIXIR Europe
    ror: 044rwnt51
    index: 2
```

Formatting

This document use Markdown and you can look at this tutorial.

Subsection level 2

Please keep sections to a maximum of only two levels.

Tables

Tables can be added in the following way, though alternatives are possible:

```
Table: Note that table caption is automatically numbered and should be given before the table itself.

| Header 1 | Header 2 |
| -------- | -------- |
| item 1 | item 2 |
| item 3 | item 4 |
```

This gives:

Table 1: Note that table caption is automatically numbered and should be given before the table itself.

Header 1	Header 2
item 1	item 2
item 3	item 4

Figures

A figure is added with:

```
![Caption for BioHackrXiv logo figure](./biohackrxiv.png)
```

This gives:

Figure 1: Caption for BioHackrXiv logo figure

Figures can be scaled by adding the width or height to the Markdown like this:

```
![Caption for BioHackrXiv logo figure](./biohackrxiv.png){ width=50px }
```

Other main section on your manuscript level 1

Lists can be added with:

1. Item 1
2. Item 2

Citation Typing Ontology annotation

You can use CiTO annotations, as explained in this BioHackathon Europe 2021 write up and this CiTO Pilot. Using this template, you can cite an article and indicate *why* you cite that article, for instance DisGeNET-RDF (Queralt-Rosinach et al., 2016).

The syntax in Markdown is as follows: a single intention annotation looks like `[@usesMethodIn:Krewinkel2017]`; two or more intentions are separated with colons, like `[@extends:discusses:Nielsen2017Scholia]`. When you cite two different articles, you use this syntax: `[@citesAsDataSource:Ammar2022ETL; @citesAsDataSource:Arend2022BioHackEU22]`.

Possible CiTO typing annotation include:

- citesAsDataSource: when you point the reader to a source of data which may explain a claim
- usesDataFrom: when you reuse somehow (and elaborate on) the data in the cited entity
- usesMethodIn
- citesAsAuthority
- citesAsEvidence
- citesAsPotentialSolution
- citesAsRecommendedReading
- citesAsRelated
- citesAsSourceDocument
- citesForInformation
- confirms
- documents
- providesDataFor
- obtainsSupportFrom
- discusses
- extends

- agreesWith
- disagreesWith
- updates
- citation: generic citation

Results

Discussion

...

Acknowledgements

...

References

Galaxy Community. (2024). The galaxy platform for accessible, reproducible, and collaborative data analyses: 2024 update. *Nucleic Acids Research*, *52*(W1), W83–W94. https://doi.org/10.1093/nar/gkae410 **[cito:extends]**

Kostrykin, L., Depoortere, B., Chiang, D., Klumpe, S., Massei, R., Kalaš, M., Fouilloux, A., & Serrano-Solano, B. (2025). *Development of FAIR image analysis workflows and training in galaxy*. https://doi.org/10.37044/osf.io/hbw6m_v1 **[cito:citesAsRelated]**

Kostrykin, L., Woller, T., Kalaš, M., Özdemir, B., Andrade Buono, R., Sun, Y., Kumar, A., Goué, N., Grüning, B., Serrano-Solano, B., & Fouilloux, A. (2024). *Enhancing the image analysis community in galaxy*. https://doi.org/10.37044/osf.io/w8dsz **[cito:citesAsRelated]**

Queralt-Rosinach, N., Pinero, J., Bravo, À., Sanz, F., & Furlong, L. I. (2016). DisGeNET-RDF: harnessing the innovative power of the Semantic Web to explore the genetic basis of diseases. *Bioinformatics*, *32*(14), 2236–2238. https://doi.org/10.1093/bioinformatics/btw214 **[cito:citesAsAuthority]**

Author contributions